

COURSE TITLE : POLYMER TECHNOLOGY
COURSE CODE : 3074
COURSE CATEGORY : B
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 60
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIOD
1	Analysis of polymers	15
2	Poly merization reactions&methods of polymerization	15
3	Manufacturing methods of plastics	15
4	Fibres,spinning methods,moulding techniques	15
TOTAL		60

COURSE OUTCOME :

SL.NO.	SUB	STUDENT WILL BE ABLE TO
1	1	Classify polymers according to commodity ,engineering and speciality
	2	Understand general properties of polymers
	3	Analyse polymer materials
2	1	Understand types of polymers and polymerisation reactions
	2	Know polymerization methods
	3	Explain manufacturing methods of different types of plastics,fibers,resins etc
3	1	Know different spinning methods and moulding techniques
	2	Know the importance of Biopolymers
	3	Know the recycling process of plastics

SPECIFIC OUTCOMES

MODULE- I

1.1.0 Know the basics polymer technology

- 1.1.1 Classify the polymers according to commodity-Engineering and speciality.
- 1.1.2 List and explain the different types of plastics, rubber, Fibers, adhesives and coatings.

1.2.0 Know the general properties of polymers.

- 1.2.1 Describe the structure, property,relationship,number average molecular weight and poly dispersity index.
- 1.2.2 Explain solubility of polymers general rules for solubility, polymer morphology.
- 1.2.3 Explain mechanical properties of polymers such as stress-strain behavior, strength, elongation, fracture and toughness.
- 1.2.4 Describe the flow behavior of polymers

1.3.0 Understand the analysis of polymer materials and the methods of manufacturing polymers

- 1.3.1 Explain melting point test, solubility test and copper wire test used for the analysis of polymer materials.
- 1.3.2 Describe the method of determination of molecular weight of polymer by viscosity method.
- 1.3.3 Describe the manufacturing method of (i) Ethylene (ii) Propylene (iii) Vinyl chloride (iv) styrene (v) Caprolactum

MODULE- II

2.1.0 Know the important polymerisation reactions.

- 2.1.1 Explain the following polymerization reactions.
 - (1) Step polymerization (2) Chain polymerization
 - (3) Anionic and cationic polymerization (4) Polymerisation by condensation

2.2.0 Differentiate the co-polymers.

- 2.2.1 Describe Random , Block and Graft type polymers.
- 2.2.2 Explain the features and application of Co-polymerisation

2.3.0 Know the different methods of polymerization

- 2.3.1 Explain the following polymerization methods.
 - (a) bulk polymerization (b) solution polymerization
 - (c) suspension polymerization (d) Emulsion polymerization
 - (e) polycondensation

2.4.0 Moulding techniques for plastics

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| a. Calendaring | b. Die casting | c. Film casting |
| d. Injection moulding | e. Blow moulding | f. Extrusion moulding |

MODULE-III

3.1.0 Know the manufacture of plastics.

- 3.1.1 Explain the manufacture of thermoplastics such as
 - (1) polyethylene (2) polypropylene
 - (3) polyvinylchloride (4) polystyrene
- 3.1.2 Describe the manufacture of thermosetting plastics such as (1) phenol formaldehyde (2) ureaformaldehyde (3) melamine formaldehyde
- 3.1.3 Explain the manufacture of polyesters, Epoxies and mention their uses.

3.2.0 Know the importance of reinforced plastics.

- 3.2.1 Describe the manufacture of reinforced plastics.
- 3.2.2 List the materials used for reinforcement mentions their effect on properties of plastics.
- 3.2.3 List the important additives used in plastics.
- 3.2.4 Explain the role of additives.

MODULE-IV

4.1.0 Know the important man-made fibers.

- 4.1.1 Differentiate natural fibers and man-made fibers.
- 4.1.2 Explain the different spinning methods such as Melt, solution and dry spinning

- 4.1.3 Describe the manufacture of following fibers and mention their uses.
(a) viscose rayon (b) polyester (c) Nylon

4.2.0 Know the importance of Biopolymers

- 4.2.1 Define natural and synthetic biopolymers and inorganic polymers
4.2.2 Explain the manufacture of any one of natural biopolymer

4.3.0 Know the recycling process of plastics

- 4.3.1 List different methods of recycling
4.3.2 Explain the method of recycling

COURSE CONTENT

MODULE-I

Introduction to polymers, Classification, General properties, Analysis of polymer materials, sources of monomers, manufacture of Ethylene, propylene, vinyl chloride, styrene, caprolactum.

MODULE-II

Types of polymerisation reaction , Types of co-polymers, Characteristics features and applications of co-polymerisation, method of polymerisation

MODULE-III

Manufacture of thermo plastics- Polyethylene, poly propylene, polyvinylchloride, polystyrene, Manufacture of thermosetting plastics-Formaldehyde based plastics, polyesters and epoxies and uses of above plastics, Reinforces plastics, additives used in plastics- moulding techniques for plastics.

MODULE-IV

Manmade fibres, Spinning methods manufacture of viscose rayon, nylon, Polyester, uses of above fibres-, biopolymers- natural-synthetic-inorganic, method of manufacture- biopolymer, plastic recycling-method of recycling and application

REFERENCES

- | | | |
|-----------------------|---|-----------------|
| Gowariker .V. R | - | Polymer Science |
| John Wiely | - | Polymer Science |
| V. Ahluwlia & Anuradh | - | Polymer Science |